

IT Assets Management System

Technical Project Report & Functional Overview

This report provides a comprehensive overview of the **IT Assets Inventory Management System**, detailing its technical foundation, architecture, security measures, and feature set.

1. Technology Stack and Languages

The system is built using a modern, scalable, and high-performance stack centered around **TypeScript** for both client and server applications.

Layer	Technologies
Frontend	Angular 17+ (Standalone), NgRx, TailwindCSS, AG Grid
Backend	NestJS 11, Node.js, TypeORM
Database	PostgreSQL (Relational Storage)
Infrastructure	Redis (Cache), MinIO (S3 Storage)
Security	JWT, Passport.js, Bcrypt

2. Database Architecture

The system utilizes **PostgreSQL**, a robust relational database management system. Data integrity and schema management are handled via **TypeORM**.

Key Entities and Structure

- **Users and Roles:** Manages system access with Role-Based Access Control (RBAC). Supports Admins, Technicians, and Employees.
- **Assets:** The core entity, tracking serial numbers, purchase dates, warranty info, and current status.
- **Categories:** Logical grouping of assets (e.g., Laptops, Networking, Monitors).
- **Locations:** Hierarchical tracking of where assets are physically located.
- **Assignments:** Tracks the historical and current link between an asset and a user/location.
- **Maintenance:** Logs for tracking repairs, service dates, and costs.

3. Security Standards and Measurements

Security is integrated at every layer of the application:

- **Authentication:** JWT (JSON Web Tokens) for stateless security and Redis-based token blacklisting for secure logouts.
- **Authorization:** Role-Based Access Control (RBAC) ensuring only authorized staff can modify sensitive data.
- **Data Integrity:** Strict DTO validation and TypeORM relationship constraints.
- **Secure Storage:** Encrypted passwords (Bcrypt) and secure S3-compatible file storage (MinIO).

4. System Architecture

The project follows a **Decoupled Client-Server Architecture**, allowing for independent scaling and maintenance of the frontend and backend.

IT ASSET MANAGEMENT SYSTEM: SOFTWARE ARCHITECTURE

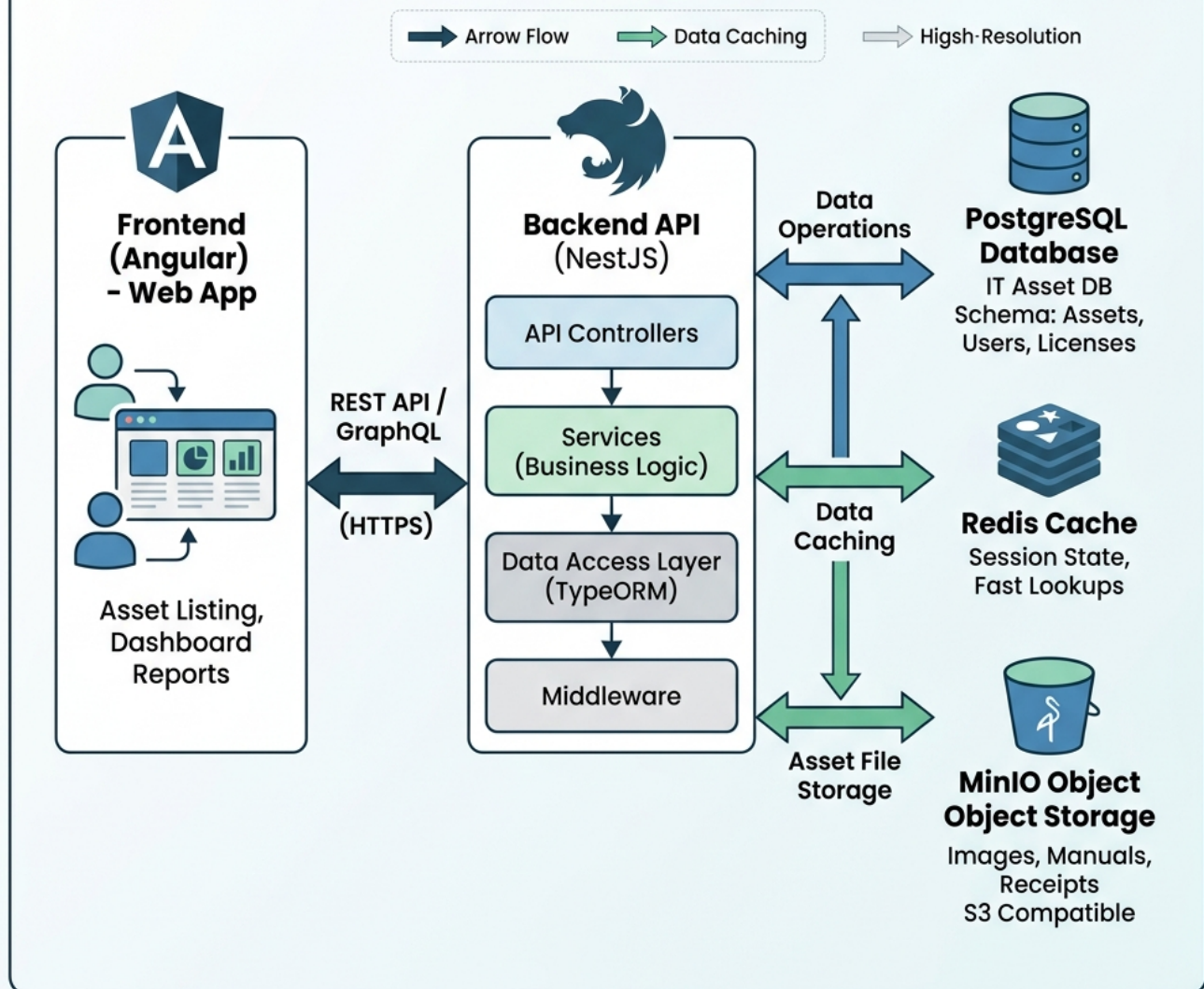


Figure 1: High-level System Architecture Diagram

5. Site Map

The navigational structure of the application is designed for efficiency, providing quick access to all management modules from a central dashboard.

IT ASSET MANAGEMENT SYSTEM - SITE MAP

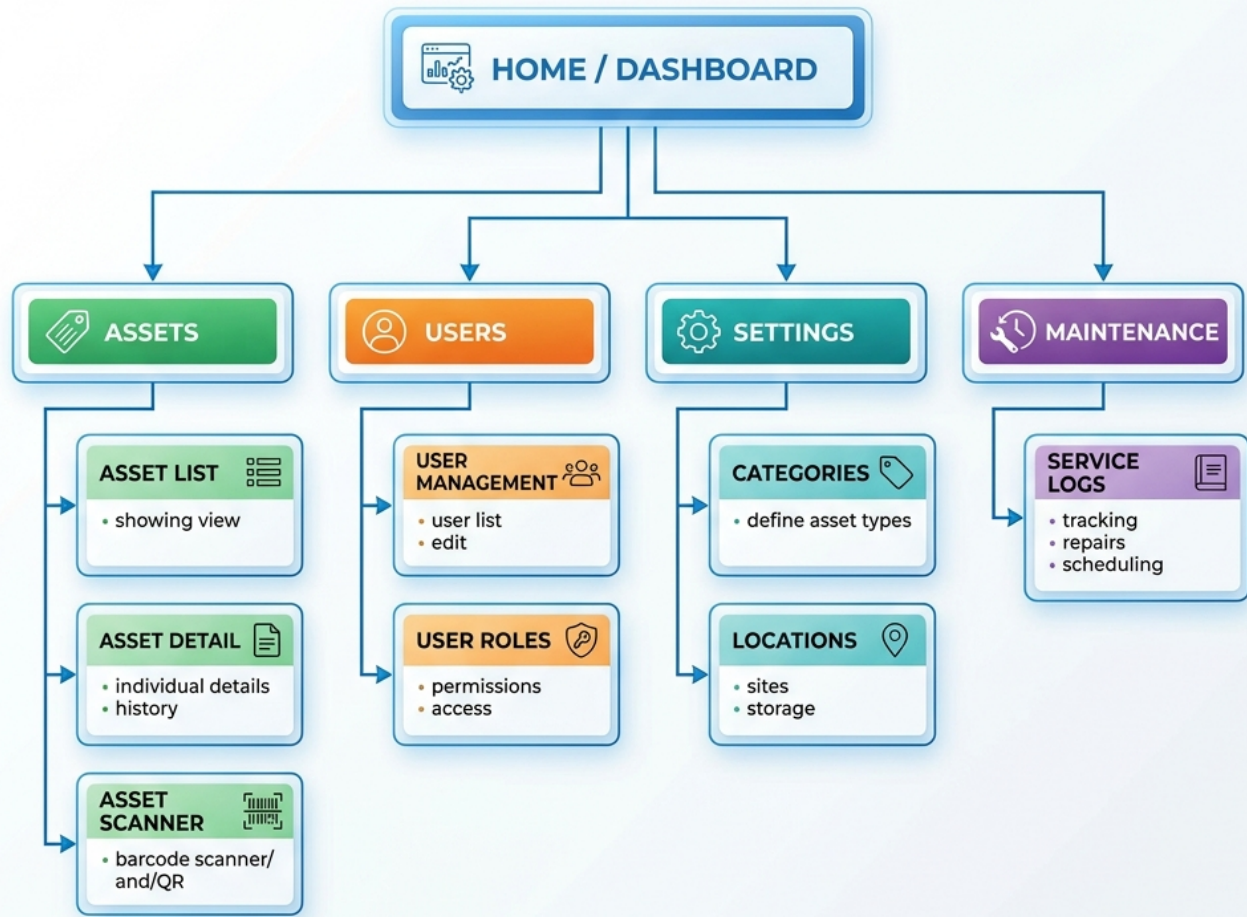


Figure 2: Application Navigational Site Map

6. Advanced Project Features

Bulk Import (Excel/JSON)

Import thousands of **Assets** and **Users** instantly using standardized templates. The system validates serial numbers and required fields automatically.

Dynamic Dashboard

Real-time visualization of asset distribution, status summaries (Available vs. Deployed), and critical maintenance alerts.

QR and Barcode Scanning

Integrated webcam scanner for instant asset lookup and rapid inventory audits directly from any device.

Asset Lifecycle

Complete tracking from procurement to retirement, including assignment history and maintenance logs.

7. System Access

Local Accessibility

For internal use within the corporate network, the system can be reached via:

- **Web Application:** `http://localhost:4200`

- **API Documentation:** <http://localhost:3000/api/docs>

Public Accessibility (Summary)

For remote access, the system can be deployed using a **Reverse Proxy (Nginx)** and **SSL (HTTPS)** to ensure secure communication via a company domain.